

DEKALB-PEACHTREE, GA, USA

WMO: 722196

Lat: 33.875N

Lon: 84.302W

Elev: 1002

StdP: 14.17

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 53863

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.2	25.7	4.9	7.3	29.0	8.7	8.9	31.8	22.6	35.5	20.4	38.2	8.4	310	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	94.1	73.5	92.0	73.4	90.2	72.9	76.9	88.0	76.0	86.5	75.2	85.0	7.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.5	129.3	79.4	73.0	127.3	79.1	72.4	124.6	78.7	40.8	88.2	40.0	87.3	39.2	85.4	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.5	14.1	DB	15.2	97.0	4.5	3.5	11.9	99.5	9.3	101.5	6.8	103.5	3.5	106.0	
			WB	13.0	78.8	4.1	1.6	10.1	79.9	7.7	80.9	5.4	81.8	2.4	83.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	42.9	47.0	53.8	62.0	70.3	77.4	79.9	79.4	73.9	63.1	52.7	45.8
	DBStd	14.92	9.91	8.72	8.60	6.98	5.96	4.08	3.46	3.70	5.29	7.47	7.82	8.65
	HDD50	735	264	149	61	5	0	0	0	0	5	60	191	
	HDD65	2807	685	505	355	135	25	0	0	0	124	375	597	
	CDD50	5273	44	65	180	366	629	821	928	913	716	411	140	60
	CDD65	1869	0	1	10	46	189	371	463	448	271	64	5	1
	CDH74	16545	1	3	87	443	1595	3386	4383	4106	2074	439	27	1
	CDH80	6559	0	0	7	73	476	1431	1930	1807	752	82	1	0
Wind	WSAvg	5.4	6.8	6.7	6.8	6.4	5.3	4.9	4.7	4.0	4.1	4.6	5.2	6.0
Precipitation	PrecAvg	51.3	4.4	4.6	5.3	3.7	3.9	4.3	4.8	4.8	4.0	3.3	4.3	4.1
	PrecMax	73.4	7.2	7.8	9.4	7.5	11.2	9.9	13.0	10.3	12.6	9.6	8.7	10.0
	PrecMin	32.6	2.3	1.8	1.9	0.7	1.3	0.9	1.4	1.9	0.2	0.2	1.0	1.1
	PrecStd	10.1	1.3	1.7	2.1	1.6	2.1	2.2	2.7	1.9	3.0	2.3	2.0	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.6	73.8	80.7	85.5	90.7	95.5	96.8	97.4	93.3	86.2	78.5	72.6
		MCWB	59.9	61.7	62.1	64.7	69.6	72.3	74.2	74.2	71.1	69.9	63.0	62.6
	2%	DB	66.0	68.7	76.3	82.0	88.0	92.8	93.6	94.0	90.3	81.8	73.0	68.5
		MCWB	57.6	58.1	60.6	64.8	69.1	72.9	74.1	74.1	70.9	66.6	61.0	61.8
	5%	DB	62.8	65.5	72.6	79.3	85.1	90.5	91.3	91.2	87.7	79.1	70.0	64.2
		MCWB	56.3	56.2	58.8	62.6	68.0	72.4	74.3	73.7	71.0	64.8	59.8	59.0
	10%	DB	58.7	62.4	69.0	75.9	82.3	88.1	89.7	89.1	84.3	75.4	66.1	60.6
		MCWB	53.1	54.0	57.1	61.3	67.4	71.8	74.0	73.4	70.2	63.3	58.0	54.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.2	64.3	65.2	69.4	73.7	77.2	78.2	78.0	75.4	73.1	69.1	67.2
		MCDB	66.3	69.2	73.2	78.9	82.5	88.7	89.1	89.6	84.4	80.9	71.4	68.8
	2%	WB	60.4	61.9	63.5	67.6	72.0	75.9	77.0	76.8	74.3	70.9	65.2	63.3
		MCDB	64.3	66.5	71.8	76.6	81.8	86.8	88.3	87.8	82.9	77.5	68.9	66.1
	5%	WB	57.6	59.1	61.7	66.0	70.8	74.8	76.1	75.8	73.3	69.0	62.8	60.3
		MCDB	61.5	63.6	69.3	74.5	80.6	85.2	86.6	86.1	81.5	74.4	66.9	63.8
	10%	WB	54.2	56.0	59.6	64.1	69.4	73.6	75.2	75.0	72.3	67.0	60.0	55.9
		MCDB	57.6	60.5	66.7	72.1	78.6	83.4	85.1	84.5	79.9	72.8	65.2	59.0
Mean Daily Temperature Range	5% DB	MDBR	18.5	19.1	20.7	21.3	20.2	18.9	18.0	18.0	18.4	20.4	20.5	17.5
		MCDBR	22.0	22.5	25.2	25.5	23.1	22.3	20.6	21.3	22.4	23.3	24.1	21.0
		MCWBR	14.7	14.0	12.2	10.6	7.9	6.8	6.0	6.1	7.4	10.4	13.2	14.5
	5% WB	MCDBR	18.1	18.2	19.9	19.6	18.5	18.3	18.0	17.8	17.0	16.8	17.5	17.6
MCWBR		14.3	14.0	11.7	10.2	7.7	7.0	6.1	6.0	6.8	9.6	12.9	14.4	
Clear-Sky Solar Irradiance	taub	0.310	0.314	0.347	0.389	0.444	0.474	0.515	0.515	0.418	0.364	0.334	0.310	
	taud	2.539	2.523	2.453	2.320	2.204	2.161	2.062	2.050	2.308	2.455	2.483	2.556	
	Ebn at Noon	288	298	295	285	269	260	248	246	268	276	275	281	
	Edh at Noon	26	29	34	41	46	48	53	52	39	31	27	24	
All-Sky Solar Radiation	RadAvg	847	1026	1373	1732	1916	1962	1875	1731	1498	1263	942	718	
	RadStd	52	128	112	113	156	156	135	98	147	156	99	75	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (40 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+265	+185

Nomenclature: See separate page